Highlights

GLP-Mamba: A Global-Local Mamba Architecture for Accurate Remote Photoplethysmography Estimation

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- GLP-Mamba enables robust remote PPG via global–local Mamba modeling.
- Twin-path Mamba captures multi-scale temporal dynamics with dual branches.
- Differential Motion Encoder fuses RGB and multi-scale motion cues.
- Experimental results outperform other methods in cross-dataset evaluations.

GLP-Mamba: A Global-Local Mamba Architecture for Accurate Remote Photoplethysmography Estimation

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Abstract

Remote photoplethysmography (rPPG) is a non-contact technique for physiological measurement from facial videos. However, accurate estimation remains challenging because pulse signals inherently contain both stable long-term rhythms and short-term local variations, the latter being extremely subtle and easily corrupted by interference. The main difficulty therefore lies in jointly capturing these multi-scale dynamics while maintaining robustness to such disturbances. To address this challenge, we propose GLP-Mamba, a global–local Mamba-based framework for robust rPPG estimation, integrating the Twin-Path Mamba (TPM) and the Differential Motion Encoder (DME). The TPM models long-term physiological rhythms through a global Mamba pathway and captures short-term local variations through a local convolutional pathway. The DME enhances the input representation by adaptively fusing multi-scale temporal difference cues with spatial appearance features, thereby suppressing transient artifacts and preserving subtle pulse-related variations. Experiments on multiple datasets demonstrate that GLP-Mamba achieves state-of-the-art performance with lower computational cost and strong cross-dataset generalization, with the largest relative improvement reaching 24.8%.

Keywords:

Remote photoplethysmography, Mamba, rPPG, Global-Local Mamba, Multi-scale Temporal Difference

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1. Introduction

Remote photoplethysmography (rPPG) has attracted increasing attention due to its ability to extract blood volume pulse (BVP) signals from facial videos in a non-contact manner. It shows great potential for applications in telemedicine and health monitoring [1, 2]. rPPG research has evolved from traditional signal processing methods to deep learning-based approaches. Early methods primarily relied on handcrafted signal processing techniques for rPPG extraction [3, 4, 5]. With the rapid development of deep learning, CNNs, Transformers, and state space models such as Mamba [6] have been increasingly introduced into rPPG estimation. Compared with traditional handcrafted approaches, recent deep learning models, especially Transformer- and Mamba-based methods, enable stronger long-range temporal modeling in rPPG estimation [7]. Despite the encouraging progress, robustly modeling the complex temporal characteristics of rPPG signals remains challenging, especially in unconstrained scenarios.

rPPG signals naturally exhibit dual temporal characteristics, with stable periodicity over long temporal windows and rapid local fluctuations and perturbations over short time scales [8]. Therefore, accurate rPPG estimation requires not only capturing global periodic rhythms over extended temporal ranges, but also perceiving short-term local dynamics. However, most existing methods typically rely on a single temporal modeling pathway [9, 10, 11, 12, 13], making it difficult to effectively balance global dependency modeling and local dynamic modeling. Methods biased toward global modeling tend to smooth short-term variations, whereas purely local modeling often lacks awareness of the overall periodic rhythm. As a result, jointly modeling and effectively fusing the global and local temporal characteristics of rPPG signals remains a key problem for accurate estimation.

Moreover, rPPG signals are inherently weak and highly susceptible to noise interference [14]. In real-world scenarios, non-physiological factors such as motion, occlusion, and illumination changes can further interfere with temporal modeling, causing subtle local pulse variations at short time scales to be easily overwhelmed and making robust rPPG estimation more challenging. Suppressing noise while preserving short-term local pulse variations therefore remains a major challenge.

To address these challenges, we propose GLP-Mamba, a global–local Mamba architecture composed of a Twin-Path Mamba (TPM) and a Differential Motion Encoder (DME). Inspired by recent advances in dual-path Mamba architectures for multi-scale feature modeling in vision tasks [15, 16, 17], we propose a TPM to address the difficulty of jointly modeling the long-range periodic rhythms and short-term local dynamics of rPPG signals. Specifically, TPM adopts a dual-path design, where a global Mamba branch captures long-range periodic dependencies and a local convolutional branch focuses on short-term temporal variations. This design directly corresponds to the global periodicity and local fluctuation characteristics of rPPG signals. To address the susceptibility of weak rPPG signals to non-physiological interference, we further introduce a DME to enhance the input representation before temporal modeling. DME adaptively fuses RGB appearance features with multi-scale temporal difference cues, thereby suppressing motion artifacts and illumination interference while preserving subtle pulse-related variations that are intrinsically tied to cardiac activity. Extensive experiments on multiple datasets demonstrate that GLP-Mamba achieves strong performance in both intra-dataset and cross-dataset evaluations.

To sum up, the main contributions of this paper are as follows:

(1) We propose GLP-Mamba, a novel global-local Mamba architecture for robust rPPG estimation. It effectively addresses the limitations of single-path temporal modeling by jointly capturing long-term periodicity and short-term dynamics.

(2) We design a TPM module that combines global Mamba for long-range dependencies and local convolution for short-term variations, enabling joint modeling of global periodic rhythms and fine-grained local temporal dynamics.

(3) We introduce a DME that fuses RGB features with multi-scale temporal differences to suppress motion artifacts while effectively capturing spatiotemporal variations.

(4) Extensive experiments demonstrate that GLP-Mamba achieves competitive performance in both intra-dataset and cross-dataset evaluations, while maintaining a clear advantage in computational cost, indicating an effective balance between efficiency and accuracy.

2. Related Work

2.1. Remote Physiological Measurement

Traditional rPPG methods recover physiological signals from subtle facial color variations in videos using handcrafted signal processing techniques, such as ICA, PCA, CHROM, and POS [3, 18, 4, 5]. However, these methods rely on idealized assumptions, and their performance often degrades substantially in complex real-world scenarios. With the advancement of deep learning, end-to-end approaches have gradually become mainstream in rPPG estimation. DeepPhys [19] first introduced deep learning to the rPPG domain, and subsequent CNN-based methods, such as PhysNet [20] and TS-CAN [21], further improved spatiotemporal representation learning. However, due to the local nature of convolution operations, these methods remain limited in capturing long-range temporal dependencies, especially in long video sequences. To address this issue, Transformer-based methods were later introduced into the rPPG domain to enhance global temporal modeling [8, 22, 23]. Although Transformers provide stronger long-range dependency modeling, their quadratic computational complexity leads to rapidly increasing costs when processing long videos, which limits their efficiency in practical applications.

More recently, state space models (SSMs) have emerged as an efficient alternative for long-sequence modeling. Among them, Mamba [6] achieves effective long-range dependency modeling with linear computational complexity, making it particularly suitable for rPPG tasks involving long facial video sequences [24]. RhythmMamba [13] was the first to introduce Mamba into the rPPG domain. MaKAN-Mixer [11] further combines Mamba with Kolmogorov-Arnold Networks (KAN) [25].

However, existing Mamba-based rPPG methods mainly focus on temporal sequence modeling within a unified representation, while the explicit decoupling of long-range rhythmic dependencies and short-term local fluctuations remains insufficiently explored. Building on these advances, GLP-Mamba introduces a task-specific global-local Mamba architecture to better match the intrinsic dual-scale temporal characteristics of rPPG signals.

2.2. Global-Local Mamba Temporal Modeling

Global-local modeling has recently been explored in Mamba-based architectures to enhance multi-scale sequence representation [26]. The core idea is to decouple temporal modeling into two complementary pathways, where the global branch emphasizes long-range dependencies and overall trends, while the local branch focuses on short-range details and rapid variations. Such a design has shown effectiveness in multiple domains, such as time series forecasting [26], multimodal medical image super-resolution [15], few-shot fault diagnosis [16], and low-light image enhancement [27].

The dual temporal characteristics of rPPG signals make global-local temporal modeling particularly suitable for this task, as accurate estimation requires jointly capturing long-range quasi-periodic rhythms and short-term local variations. Based on this observation, the proposed GLP-Mamba is designed as a dedicated global-local Mamba architecture for rPPG estimation.

3. Methodology

3.1. Overall Architecture

We propose GLP-Mamba, a global–local Mamba framework for robust rPPG estimation. As illustrated in Figure 1, the model consists of four main components: a DME for multi-scale temporal feature encoding, a Spatio-Temporal Aggregation module for compact feature compression, a stack of Temporal Mamba Blocks for temporal modeling, and a signal reconstruction head for final rPPG waveform prediction.

Given an input video $\mathbf{X} \in \mathbb{R}^{B\times D\times 3\times H\times W}$, where B denotes the batch size, D is the number of frames, and $H \times W$ represents the spatial resolution, the model outputs a predicted signal $\hat{\mathbf{y}} \in \mathbb{R}^{B\times D}$. The input video is first processed by the DME to extract frame-wise representations that encode both RGB appearance information and multi-scale temporal difference cues. To reduce computational complexity while retaining essential temporal information, a Spatio-Temporal Aggregation module is introduced to compress the feature sequence into a compact representation. The aggregated features are then fed into a series of Temporal Mamba Blocks for hierarchical temporal modeling. Each block contains a TPM-based temporal modeling sub-layer and a Frequency-Domain FFN sub-layer. In the TPM sub-layer, the input sequence is processed through a twin-path design, where a global Mamba branch captures long-range periodic dependencies and a local multi-scale temporal convolution branch models short-term variations. Finally, the refined temporal features are upsampled and projected to reconstruct the one-dimensional rPPG signal corresponding to the input video frames.

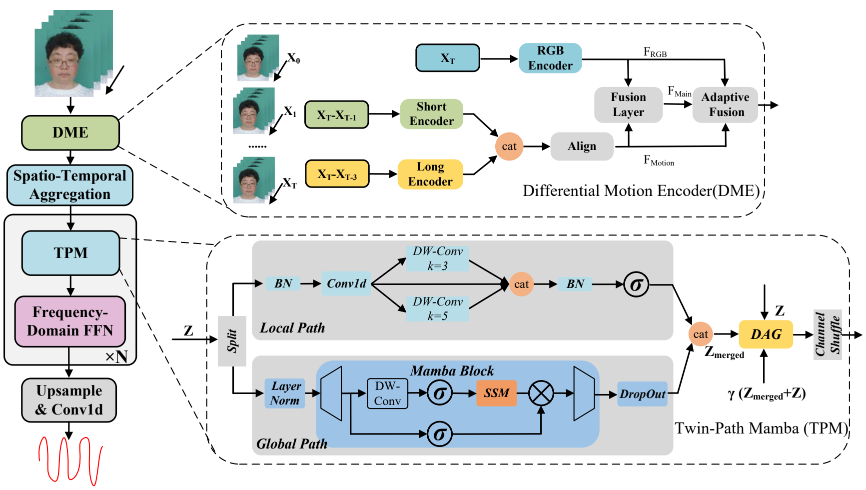


Figure 1: Overall framework of the proposed GLP-Mamba. σ represents the activation function, and the trapezoid denotes the linear layer. In the DME module, the first frame $\mathbf{X}_0$ is used for boundary padding when $t-1<0$ or $t-3<0$. In the TPM module, $\mathbf{Z}$ denotes the input sequence, and $\mathbf{Z}_{\text{merged}}$ denotes the concatenated output of the global and local branches. DAG denotes the Domain-Adaptive Gating mechanism.

3.2. *Differential Motion Encoder*

The quality of temporal modeling in Mamba largely depends on the informativeness of the input representation. Existing rPPG methods typically rely on either raw RGB videos [28, 29] or temporal differences [19, 21, 13], which emphasize appearance and motion cues, respectively. To leverage both appearance and motion information, we design the DME to extract RGB features together with short-term and long-term temporal difference features, and then integrate them through adaptive residual weighting to produce spatiotemporally coupled representations for subsequent temporal modeling.

Multi-scale Temporal Difference and Feature Encoding. Pulse-related temporal variations may manifest differently across short and long temporal spans. To capture both instantaneous changes and longer-range motion patterns, we construct two temporal difference sequences from the input video $\mathbf{X} \in \mathbb{R}^{B \times D \times 3 \times H \times W}$, where B denotes the batch size, D is the number of frames, and $H \times W$ represents the spatial resolution. At each time step $t \in [0, D-1]$, the short-term and long-term differences are defined as:

$$\begin{aligned} \mathbf{X}_{\text{short}}(t) &= \mathbf{X}(t) - \mathbf{X}(t-1), \\ \mathbf{X}_{\text{long}}(t) &= \mathbf{X}(t) - \mathbf{X}(t-3). \end{aligned} \tag{1}$$

For the initial frames, when $t-1 < 0$ or $t-3 < 0$, we use the first frame $\mathbf{X}(0)$ as a substitute so that the generated difference sequences have the same length as the original input. This boundary strategy allows DME to obtain valid frame-difference responses from the beginning of the input sequence without delaying the temporal response. These temporal difference operations suppress static backgrounds and slowly varying illumination changes, thereby enhancing motion patterns relevant to physiological activity.

The RGB sequence and the two temporal difference sequences are then reshaped into $(B \cdot D)$ frame-level samples for efficient feature extraction with 2D convolutions. The DME uses three parallel convolutional branches to encode complementary appearance and motion features. All three branches use lightweight 2D convolutional stems with a 7×7 convolution, batch normalization, ReLU activation, and 2×2 max pooling to produce frame-level features at a spatial resolution of $H/4 \times W/4$. In the RGB branch, an additional dropout layer is used after activation. The RGB branch extracts $d/2$ channels, while the short- and long-difference branches each extract $d/4$ channels, where d is the output dimension of the DME. The three branch features are formulated as:

$$
\begin{aligned}
\mathbf{F}_{\text{rgb}} &= \text{Conv}_{\text{rgb}}(\mathbf{X}) \in \mathbb{R}^{(BD)\times(d/2)\times(H/4)\times(W/4)}, \\
\mathbf{F}_{\text{short}} &= \text{Conv}_{\text{short}}(\mathbf{X}_{\text{short}}) \in \mathbb{R}^{(BD)\times(d/4)\times(H/4)\times(W/4)}, \\
\mathbf{F}_{\text{long}} &= \text{Conv}_{\text{long}}(\mathbf{X}_{\text{long}}) \in \mathbb{R}^{(BD)\times(d/4)\times(H/4)\times(W/4)}.
\end{aligned} \tag{2}
$$

To facilitate subsequent fusion, the two motion features are concatenated along the channel dimension and aligned to $d/2$ channels through a 1×1 convolution followed by batch normalization:

$$
\mathbf{F}_{\text{motion}} = \text{BN}(\text{Conv}_{1\times1}([\mathbf{F}_{\text{short}}, \mathbf{F}_{\text{long}}])) \in \mathbb{R}^{(BD)\times(d/2)\times(H/4)\times(W/4)}, \tag{3}
$$

where $[\cdot, \cdot]$ denotes channel-wise concatenation. This design allows short-term and long-term motion features to be integrated before fusion with the appearance branch. At this stage, we obtain two feature representations with matched dimensions: the appearance feature $\mathbf{F}_{\text{rgb}}$ and the motion feature $\mathbf{F}_{\text{motion}}$.

Adaptive Feature Fusion. The appearance feature **F**rgb and the aligned motion feature **F**motion are concatenated along the channel dimension and passed through a fusion pathway consisting of a 5×5 convolution, batch normalization, ReLU activation, dropout, and 2×2 max pooling. This further reduces the spatial resolution to $H/8 \times W/8$ and yields the backbone feature $\mathbf{F}_{\text{main}} \in \mathbb{R}^{(BD)\times d\times(H/8)\times(W/8)}$.

To adaptively balance appearance and motion information, we introduce a residual weighting mechanism conditioned on global context. Specifically, global average pooling is first applied to the concatenated features, followed by a two-layer fully connected network and softmax normalization to predict adaptive weights $\mathbf{w} = [w_1, w_2]$ for the appearance and motion branches. Meanwhile, the two branch features are projected to d channels through 1×1 convolutions and downsampled to the same spatial resolution as the main pathway. The final DME output is obtained by weighted residual fusion:

$$
\mathbf{F}_{\text{DME}} = \mathbf{F}_{\text{main}} + \sigma(\alpha) \cdot w_1 \cdot \mathcal{P}(\mathbf{F}_{\text{rgb}}) + \sigma(\beta) \cdot w_2 \cdot \mathcal{P}(\mathbf{F}_{\text{motion}}), \tag{4}
$$

where $\mathcal{P}(\cdot)$ denotes the projection and downsampling operations, and α and β are learnable scalar parameters constrained to $(0, 1)$ by the sigmoid function $\sigma(\cdot)$ for stable training. This mechanism enables the DME to dynamically reweight appearance and motion features, producing more robust representations for subsequent temporal modeling.

3.3. Spatio-Temporal Aggregation

To bridge the DME and the subsequent temporal modeling stage with low computational overhead, we introduce a spatiotemporal aggregation layer. First, the frame-wise features $\mathbf{F}_{\text{DME}} \in \mathbb{R}^{(BD)\times d\times(H/8)\times(W/8)}$ are reshaped back into the video format $\mathbb{R}^{B\times d\times D\times(H/8)\times(W/8)}$. A lightweight 3D convolution with kernel size $(2, 5, 5)$, stride $(2, 1, 1)$, and padding $(0, 2, 2)$ is applied to aggregate local spatiotemporal patterns, reducing the temporal length to $T = D/2$ while projecting the channels from d to C, yielding $\mathbf{U} \in \mathbb{R}^{B\times C\times T\times(H/8)\times(W/8)}$. We then generate a spatial attention mask by applying a sigmoid function followed by spatial normalization, and use it to reweight $\mathbf{U}$. Finally, spatial mean pooling is applied to obtain a compact sequence representation. This is rearranged into $\mathbf{Z} \in \mathbb{R}^{B\times T\times C}$ to serve as the input for the subsequent Temporal Mamba Blocks.

3.4. Temporal Mamba Block

After the DME and the subsequent spatio-temporal aggregation, we obtain a compact sequence representation $\mathbf{Z} \in \mathbb{R}^{B\times T\times C}$ for temporal modeling. Each Temporal Mamba Block follows a residual pre-normalization design. It first performs temporal modeling through a TPM-based multi-temporal processing strategy, and then applies a Frequency-Domain FFN sub-layer [13] for feature transformation. The TPM is the core component of the block, which jointly models long-range periodic rhythms and short-term local dynamics through a dual-path architecture.

Twin-Path Mamba. Given the input sequence $\mathbf{Z} \in \mathbb{R}^{B\times T\times C}$, the TPM first partitions it into a global branch and a local branch along the channel dimension, where the global branch occupies approximately half of the channels and is aligned to a multiple of 8 for implementation efficiency:

$$\mathbf{Z}_{\text{global}} = \mathbf{Z}[:, :, : C/2], \quad \mathbf{Z}_{\text{local}} = \mathbf{Z}[:, :, C/2 :]. \tag{5}$$

The global branch captures long-range periodic dependencies using the Mamba state space model:

$$\mathbf{Z}_{\text{global}}^{\text{out}} = \text{Mamba}(\text{LN}(\mathbf{Z}_{\text{global}})), \tag{6}$$

while the local branch models short-term temporal variations by first applying channel reduction with a 1×1 temporal projection, followed by depthwise temporal convolutions with kernel sizes 3 and 5, and finally projecting the concatenated local features back to the original local dimension:

$$\mathbf{Z}_{\text{local}}^{\text{out}} = \text{Linear}(\text{SiLU}(\text{Concat}(\mathbf{F}_{\text{reduced}}, \mathbf{F}_3, \mathbf{F}_5))), \tag{7}$$

where $\mathbf{F}_{\text{reduced}}$ denotes the dimension-reduced feature obtained by a 1×1 convolution, and $\mathbf{F}_3$ and $\mathbf{F}_5$ are the outputs of depthwise convolutions with kernel sizes 3 and 5, respectively.

To improve robustness under distribution shifts, we introduce a lightweight Domain-Adaptive Gating (DAG) mechanism. The outputs of the global Mamba branch and the local temporal convolution branch are first concatenated as

$$\mathbf{Z}_{\text{merged}} = \text{Concat}(\mathbf{Z}_{\text{global}}^{\text{out}}, \mathbf{Z}_{\text{local}}^{\text{out}}) \in \mathbb{R}^{B \times T \times C}. \tag{8}$$

Given the input sequence $\mathbf{Z} \in \mathbb{R}^{B \times T \times C}$, DAG computes the channel-wise batch mean $\boldsymbol{\mu}_B$ and variance $\boldsymbol{\nu}_B$ over the batch and temporal dimensions. These statistics are compared with the running mean $\bar{\boldsymbol{\mu}}$ and running variance $\bar{\boldsymbol{\nu}}$ maintained during training with a momentum of 0.1. The normalized mean and variance deviations are computed as $\Delta_\mu = (\boldsymbol{\mu}_B - \bar{\boldsymbol{\mu}})/(\sqrt{\bar{\boldsymbol{\nu}}} + \epsilon)$ and $\Delta_\nu = (\boldsymbol{\nu}_B - \bar{\boldsymbol{\nu}})/(\bar{\boldsymbol{\nu}} + \epsilon)$, where $\epsilon = 10^{-5}$. These deviations are then fed into a lightweight gating network $G(\cdot)$ to produce a bounded channel-wise adjustment vector:

$$\mathbf{r} = 1 + \text{clip}\Big(G([\Delta_\mu, \Delta_\nu]), -0.15, 0.15\Big), \quad \mathbf{r} \in [0.85, 1.15]^C. \tag{9}$$

Here, $G(\cdot)$ is implemented as a two-layer MLP with LayerNorm, ReLU, Dropout, and Tanh activation. This bounded adjustment prevents unstable fusion while allowing the model to adapt to feature distribution shifts.

The adjustment vector $\mathbf{r}$ is used to modulate the fusion weights of three feature paths, including the merged global–local feature, the input residual, and their residual interaction. Let $\boldsymbol{\theta} = [\theta_m, \theta_z, \theta_r]$ be a learnable fusion parameter vector, corresponding to the three feature paths. The base fusion weights are obtained by $[a_m, a_z, a_r] = \mathrm{Softmax}(\boldsymbol{\theta})$. The DAG-modulated fusion weights are computed as $\boldsymbol{\omega}_m = a_m\mathbf{r}/\mathbf{s}$, $\boldsymbol{\omega}_z = a_z(2 - \mathbf{r})/\mathbf{s}$, and $\boldsymbol{\omega}_r = a_r/\mathbf{s}$, where $\mathbf{s} = a_m\mathbf{r} + a_z(2 - \mathbf{r}) + a_r$ is the channel-wise normalization factor. The fused TPM output is formulated as

$$\mathbf{Z}_{\text{fused}} = \boldsymbol{\omega}_m \odot \mathbf{Z}_{\text{merged}} + \boldsymbol{\omega}_z \odot \mathbf{Z} + \boldsymbol{\omega}_r \odot \gamma(\mathbf{Z}_{\text{merged}} + \mathbf{Z}), \tag{10}$$

where $\odot$ denotes channel-wise multiplication and γ is a learnable residual scaling parameter.

Finally, a lightweight channel shuffle operation is applied to $\mathbf{Z}_{\text{fused}}$ to promote cross-branch feature interaction, yielding the final TPM output.

Block Formulation and Stacking. For the n-th Temporal Mamba Block, let $\mathbf{Z}^{(n)} \in \mathbb{R}^{B\times T\times C}$ denote its input. The block first performs multi-temporal parallel temporal modeling by constructing several shifted sequence copies and processing them jointly through the TPM, and then applies a Frequency-Domain FFN for feature transformation. Formally,

$$\hat{\mathbf{Z}}^{(n)} = \mathbf{Z}^{(n)} + \mathcal{T}\Big(\mathbf{Z}^{(n)}\Big), \tag{11}$$

$$\mathbf{Z}^{(n+1)} = \hat{\mathbf{Z}}^{(n)} + \mathrm{FFN}\Big(\mathrm{LN}\Big(\hat{\mathbf{Z}}^{(n)}\Big)\Big), \tag{12}$$

where $\mathcal{T}(\cdot)$ denotes the complete TPM-based temporal modeling operation defined above, including the global Mamba branch, the local temporal convolution branch, DAG fusion, and channel shuffle, and $\mathrm{FFN}(\cdot)$ denotes the Frequency-Domain FFN [13].

We stack N Temporal Mamba Blocks to progressively model long-range temporal dependencies:

$$\mathbf{Z}^{(N)} = \mathcal{B}^{(N-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ \mathcal{B}^{(0)}(\mathbf{Z}^{(0)}), \tag{13}$$

where $\mathcal{B}^{(n)}(\cdot)$ denotes the n-th Temporal Mamba Block and $\mathbf{Z}^{(0)} = \mathbf{Z}$. In our implementation, the stacking depth is set to $N = 24$.

The final representation $\mathbf{Z}^{(N)} \in \mathbb{R}^{B\times T\times C}$ is transposed to channel-first format and upsampled from $T = D/2$ back to the original temporal length D, followed by a 1×1

temporal convolution to generate the predicted rPPG signal $\hat{\mathbf{y}} \in \mathbb{R}^{B \times D}$.

3.5. Training Objective

We employ a hybrid temporal-frequency loss function [13, 30] for end-to-end training. The overall loss is formulated as

$$L_{\text{total}} = \lambda_{time} L_{\text{time}} + \lambda_{freq} L_{\text{freq}}, \tag{14}$$

where the loss weights are set to $\lambda_{time} = 0.2$ and $\lambda_{freq} = 1.0$, following the temporal-frequency supervision setting in [13, 30].

The temporal loss L_{time} is defined as the negative Pearson correlation coefficient between the predicted rPPG signal $\hat{S}$ and the ground-truth signal S_{gt}, where T denotes the signal length:

$$L_{\text{time}} = 1 - \frac{\sum_{i=1}^{T} \left(\hat{S}_i - \bar{\hat{S}}\right)\left(S_{gt,i} - \bar{S}_{gt}\right)}{\sqrt{\sum_{i=1}^{T} \left(\hat{S}_i - \bar{\hat{S}}\right)^2} \sqrt{\sum_{i=1}^{T} \left(S_{gt,i} - \bar{S}_{gt}\right)^2}}. \tag{15}$$

The frequency loss L_{freq} is computed as the cross-entropy loss between the PSD response of the predicted signal and the ground-truth HR index:

$$L_{\text{freq}} = \text{CE}\Big(\text{PSD}(\hat{S}), k^*\Big), \quad k^* = \arg\max_k \text{PSD}(S_{gt})_k, \tag{16}$$

where $\text{PSD}(\cdot)$ denotes the power spectral density, and k^* is the dominant HR index derived from the ground-truth signal. In our implementation, the PSD response is computed via FFT over the BPM range of $[45, 150)$ and normalized by the sum of all frequency responses within this range before the cross-entropy computation.

4. Experiments Settings

4.1. Datasets and Metrics

We evaluate the proposed method on three public benchmark datasets: PURE [31], UBFC-rPPG [32], and MMPD [33]. Compared with PURE and UBFC-rPPG, MMPD is a relatively larger-scale and more challenging dataset, with broader variations in skin tone, motion, and lighting conditions.

PURE: Contains 60 one-minute videos from 10 subjects recorded under six controlled motion scenarios, including steady, talking, translation, and rotation, at 640×480 pixels and 30 fps. Ground-truth pulse signals were synchronously recorded using a finger-clip pulse oximeter.

UBFC-rPPG: Contains 42 videos from 42 subjects recorded in a stationary setting at 640×480 resolution and 30 fps, each lasting about 60 seconds. Ground-truth PPG signals were synchronously collected using a transmissive pulse oximeter.

MMPD: Collected using mobile devices from 33 participants with diverse skin tones across various lighting and motion conditions. We use the mini-MMPD version for comparability with existing work.

To comprehensively evaluate model performance, we adopt five standard metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Pearson correlation coefficient (ρ), and Signal-to-Noise Ratio (SNR).

4.2. Implementation Details

We establish a unified evaluation platform based on the PyTorch framework and the open-source rPPG toolbox [34], ensuring that all comparison methods are implemented under the same code framework for fair comparison. The preprocessing pipeline is as follows. Face detection is performed on the first frame using the OpenCV Haar Cascade detector with an enlarged bounding box (coefficient 1.5), and the same bounding box is applied to all subsequent frames to avoid temporal jitter. The cropped face region is resized to 128×128 pixels and z-score standardized. Each video is segmented into non-overlapping clips of 160 frames at 30 Hz. All experiments were conducted on a workstation equipped with an NVIDIA RTX 3090 GPU. The model was trained for 30 epochs with a batch size of 8 and an initial learning rate of 1.5×10^{-4}. For reproducibility, the random seed is fixed to 100 for all random number generators. The key hyperparameters are summarized in Table 1, where α, β, and γ denote learnable parameters and the listed values indicate their initial values.

Table 1: Key hyperparameter settings.

Hyperparameter	Symbol	Value
Number of GLP-Mamba blocks	N	24
Channel dimension	C	96
MLP expansion ratio	–	2
Stochastic depth rate	–	0.1
Learnable DME residual weights (init.)	α, β	0.5, 0.5
Learnable residual scaling (init.)	γ	0.5
Momentum for running statistics in DAG	–	0.1
Loss weights	$\lambda_{time}, \lambda_{freq}$	0.2, 1.0
Random seed	–	100

5. Results and Analyses

5.1. Intra-Dataset Evaluation

To validate the effectiveness of the proposed model, intra-dataset experiments were conducted on the PURE [31], UBFC-rPPG [32], and MMPD [33] datasets.

Evaluation settings for PURE and UBFC-rPPG. For the PURE dataset, we follow the split protocol in [35, 36], using the first 60% of samples in the original order for training and the remaining 40% for testing. For the UBFC-rPPG dataset, we follow the setting in [37, 36], using the first 30 samples for training and the next 12 samples for testing.

Evaluation settings for MMPD. To evaluate the method on a large-scale dataset, following [13, 14], the MMPD dataset was split in chronological order into training, validation, and testing sets with a ratio of 7:1:2, which is consistent with existing works adopting the same protocol.

Table 2 presents the intra-dataset evaluation results on the PURE, UBFC-rPPG, and MMPD datasets.

Overall, the proposed method achieves strong performance across the three benchmark datasets. Compared with representative CNN-based methods, such as DeepPhys [19] and PhysNet [20], and Transformer-based methods, such as PhysFormer [28], our method achieves lower MAE and RMSE across the three datasets. This demonstrates that the proposed GLP-Mamba, built upon the Mamba architecture, is more

Table 2: Intra-dataset evaluation results on PURE, UBFC-rPPG, and MMPD. The best results are highlighted in **bold**, and the second-best results are <u>underlined</u>. In case of ties, only our method is emphasized for clarity.

Method	PURE			UBFC-rPPG			MMPD		
	MAE↓	RMSE↓	ρ↑	MAE↓	RMSE↓	ρ↑	MAE↓	RMSE↓	ρ↑
HR-CNN [35]	1.84	2.37	0.98	4.90	5.89	0.64	-	-	-
SynRhythm [38]	2.71	4.86	0.98	5.59	6.82	0.72	-	-	-
DeepPhys [19]	0.83	1.54	0.99	6.27	10.82	0.65	22.27	28.92	-0.03
PhysNet [20]	2.10	2.60	0.99	2.95	3.67	0.97	4.80	11.80	0.60
Meta-rPPG [39]	2.52	4.63	0.98	5.97	7.42	0.53	-	-	-
TS-CAN [21]	2.48	9.01	0.92	1.70	2.72	0.99	9.71	17.22	0.44
DeeprPPG [40]	0.28	0.43	0.99	0.67	1.70	0.99	-	-	-
PulseGAN [37]	2.28	4.29	0.99	1.19	2.10	0.98	-	-	-
Dual-GAN [36]	0.82	1.31	0.99	**0.44**	0.67	0.99	-	-	-
PhysFormer [28]	1.10	1.75	0.99	0.50	0.71	0.99	11.99	18.41	0.18
ETA-rPPGNet [10]	0.34	0.77	0.99	1.46	3.97	0.93	-	-	-
TFA-PFE [41]	1.44	2.50	-	0.76	1.62	-	-	-	-
SINC [42]	0.61	1.84	0.99	0.59	1.83	0.99	-	-	-
EfficientPhys [9]	1.33	5.99	0.97	1.14	1.81	0.99	13.47	21.32	0.21
PhysMamba [7]	1.10	1.75	0.99	0.50	0.71	0.99	–	–	–
RhythmMamba [13]	0.23	0.34	0.99	0.50	0.75	0.99	3.16	7.27	0.84
MaKAN-Mixer [11]	0.25	0.40	0.99	<u>0.45</u>	<u>0.66</u>	0.99	3.09	<u>6.25</u>	**0.89**
DualPhysMamba [12]	<u>0.18</u>	<u>0.31</u>	0.99	<u>0.45</u>	0.73	0.99	**2.83**	**6.17**	0.89
Ours	**0.15**	**0.29**	**0.99**	0.49	**0.64**	**0.99**	<u>3.08</u>	6.50	<u>0.88</u>

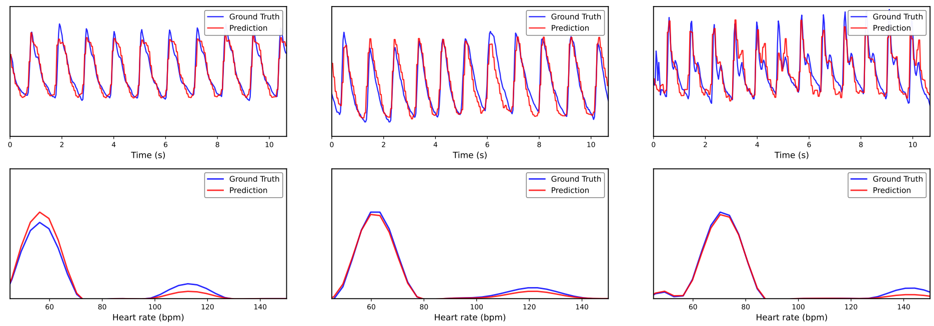
effective in capturing the temporal characteristics of rPPG signals than CNN- and Transformer-based approaches. Compared with recent Mamba-based baselines, including RhythmMamba [13], MaKAN-Mixer [11], and DualPhysMamba [12], our method achieves lower errors and comparable or better correlation performance across multiple datasets. In particular, on the PURE dataset, our method reduces the MAE from 0.18 to 0.15 and the RMSE from 0.31 to 0.29 compared with the second-best results, corresponding to improvements of 16.7% and 6.5%, respectively. On the UBFC-rPPG dataset, our method achieves the lowest RMSE of 0.64 bpm, indicating more stable prediction performance. These improvements demonstrate the effectiveness of the proposed global–local Mamba architecture. Notably, on the more challenging MMPD dataset, our method ranks second with an MAE of 3.08 bpm, remaining within 8.8% of the best result. This may be attributed to the more complex scenarios in MMPD, including stronger motion interference, more diverse skin tones, and larger illumination variations, posing greater challenges that highlight a direction for future improvement. These results collectively demonstrate the effectiveness of the proposed GLP-Mamba for robust rPPG estimation.

Additionally, we visualize the signal reconstruction quality by showing predicted pulse signals in the time and frequency domains. As shown in Fig. 2, the predicted signals exhibit strong agreement with the ground truth, accurately capturing the characteristic peaks and variations of the original rPPG waveform. This further demonstrates the effectiveness of our model in rPPG signal reconstruction.

5.2. Cross-Dataset Evaluation

To evaluate the cross-dataset generalization capability of the model, we follow the protocol in [34] for cross-dataset evaluation. Specifically, we train models on PURE and UBFC-rPPG datasets separately and then test on the other datasets. For each training dataset, we use the first 80% for training and the remaining 20% for validation. This setup produces 4 cross-dataset evaluation combinations: training on PURE and testing on UBFC-rPPG and MMPD, as well as training on UBFC-rPPG and testing on PURE and MMPD.

Cross-dataset evaluation is a key indicator for assessing model generalization ca-



(a) PURE (b) UBFC-rPPG (c) MMPD

Figure 2: Visualization of rPPG predictions on PURE, UBFC-rPPG, and MMPD datasets. Each column shows time-domain signal comparisons (top) and power spectral density comparisons (bottom), demonstrating waveform consistency and dominant frequency accuracy across datasets.

pability. Tables 3, 4, and 5 presents the cross-dataset testing results on PURE, UBFC-rPPG, and MMPD datasets. We adopt two training configurations and compare with both traditional signal processing methods and deep learning approaches. Overall, our method achieves superior performance and outperforms most advanced methods across different train–test combinations.

When training on UBFC-rPPG and testing on PURE (Table 3), our method achieves the best performance across all evaluation metrics, including MAE, RMSE, MAPE, ρ, and SNR. Compared with DualPhysMamba [12], our method reduces the MAE, RMSE, and MAPE from 0.63 to 0.54 bpm, from 1.13 to 0.85 bpm, and from 0.93 to 0.81, corresponding to improvements of 14.3%, 24.8%, and 12.9%, respectively. When training on PURE and testing on UBFC-rPPG (Table 4), our method again achieves the best overall performance across evaluation metrics. In particular, it achieves an MAE of 0.77 bpm, an RMSE of 1.13 bpm, and a MAPE of 0.83, with the RMSE showing a 19.9% improvement over the second-best result. On the more challenging MMPD dataset (Table 5), our method maintains robust performance. When training on UBFC-rPPG and testing on MMPD, our method outperforms advanced methods such as MaKAN-Mixer [11] and achieves the best overall performance among the compared methods. Compared with the second-best RhythmMamba [13], our method reduces the MAE, RMSE, and MAPE by 8.8%, 4.8%, and 3.1%, respec-

Table 3: Cross-dataset results for models trained on UBFC-rPPG and tested on PURE.

Method	MAE↓	RMSE↓	MAPE↓	ρ↑	SNR↑
GREEN [43]	10.09	23.85	10.28	0.34	-2.66
ICA [3]	4.77	16.07	4.47	0.72	5.24
CHROM [4]	5.77	14.93	11.52	0.81	4.58
LGI [44]	4.61	15.38	4.96	0.77	4.50
PBV [45]	3.92	12.99	4.84	0.84	2.30
POS [5]	3.67	11.82	7.25	0.88	6.87
DeepPhys [19]	5.54	18.51	5.32	0.66	4.40
PhysNet [20]	8.06	19.71	13.67	0.61	6.68
TS-CAN [21]	3.69	13.80	3.39	0.82	5.26
PhysFormer [28]	12.92	24.36	23.92	0.47	2.16
EfficientPhys [9]	5.47	17.04	5.40	0.71	4.09
Spiking-Phys [46]	3.83	—	5.70	0.83	—
RhythmMamba [13]	1.98	6.51	3.59	0.96	8.94
MaKAN-Mixer [11]	0.71	1.62	—	0.99	—
DualPhysMamba [12]	<u>0.63</u>	<u>1.13</u>	<u>0.93</u>	0.99	<u>10.84</u>
Ours	**0.54**	**0.85**	**0.81**	**0.99**	**10.88**

tively, while improving the SNR from -8.28 to -7.37. When training on PURE and testing on MMPD, GLP-Mamba also remains competitive, achieving the best MAE of 9.56 bpm and the best MAPE of 11.06. These results further demonstrate the strong cross-dataset generalization capability of our method.

To further evaluate the proposed method under dataset shift, Fig. 3 illustrates the Bland-Altman (BA) and scatter plots for cross-dataset scenarios. As shown, the BA plots demonstrate small mean biases and compact limits of agreement, while the scatter plots reveal a strong linear correlation with points tightly clustered along the diagonal line $y = x$, indicating robust generalization.

To further evaluate the cross-dataset generalization ability under a multi-source setting, we additionally conduct a dual-source training and single-target testing experiment, where PURE and UBFC-rPPG are jointly used for training and MMPD serves as the unseen target domain for testing only. As shown in Table 6, our method achieves the lowest MAE and the highest correlation coefficient among all compared methods.

Table 4: Cross-dataset results for models trained on PURE and tested on UBFC-rPPG.

Method	MAE↓	RMSE↓	MAPE↓	$\rho\uparrow$	SNR↑
GREEN [43]	19.73	31.00	18.72	0.37	-11.18
ICA [3]	16.00	25.65	15.35	0.44	-9.91
CHROM [4]	4.06	8.83	3.84	0.89	-2.96
LGI [44]	15.80	28.55	14.70	0.36	-8.15
PBV [45]	15.90	26.40	15.17	0.48	-9.16
POS [5]	4.08	7.72	3.93	0.92	-2.39
DeepPhys [19]	1.21	2.90	1.42	0.99	1.74
PhysNet [20]	0.98	2.48	1.12	0.99	1.09
TS-CAN [21]	1.30	2.87	1.50	0.99	1.49
PhysFormer [28]	1.44	3.77	1.66	0.98	0.18
EfficientPhys [9]	2.07	6.32	2.10	0.94	-0.12
Spiking-Phys [46]	2.80	—	2.81	0.95	—
RhythmMamba [13]	0.95	1.83	1.04	0.99	6.35
MaKAN-Mixer [11]	<u>0.77</u>	<u>1.41</u>	—	0.99	—
DualPhysMamba [12]	0.86	1.60	<u>0.89</u>	0.99	<u>6.86</u>
Ours	**0.77**	**1.13**	**0.83**	**0.99**	**6.92**

Compared with PhysLLM [47] and PHASE-Net [48], GLP-Mamba reduces the MAE from 9.95 and 9.76 to 9.62 bpm, corresponding to relative improvements of 3.3% and 1.4%, respectively. In addition, compared with PHASE-Net, our method improves the correlation coefficient from 0.39 to 0.42, indicating a stronger agreement between the estimated and ground-truth heart rates on the unseen MMPD dataset. Although PhysLLM obtains the lowest RMSE, our method achieves the best MAE and correlation performance, demonstrating its strong robustness and generalization capability under dual-source cross-dataset evaluation.

The results in both the intra-dataset and cross-dataset evaluations underscore the effectiveness of GLP-Mamba, and the additional dual-source evaluation further confirms its robustness under multi-source training. In particular, the strong cross-dataset performance may be largely attributed to the DME, which enhances subtle pulse-related temporal dynamics while suppressing dataset-specific appearance interference. Furthermore, the TPM effectively captures both long-range physiological rhythms and

Table 5: Cross-dataset results for models trained on UBFC-rPPG and PURE and tested on MMPD.

Method	Train set	MMPD				
		MAE↓	RMSE↓	MAPE↓	ρ ↑	SNR↑
GREEN [43]	—	21.73	27.72	24.44	-0.02	-14.34
ICA [3]	—	18.57	24.28	20.85	0.00	-13.84
CHROM [4]	—	13.63	18.75	15.96	0.08	-11.74
LGI [44]	—	17.02	23.28	18.92	0.04	-13.15
PBV [45]	—	17.88	23.53	20.11	0.09	-13.88
POS [5]	—	12.34	17.70	14.43	0.17	-11.53
DeepPhys [19]	UBFC	17.50	25.00	19.27	0.05	-11.72
	PURE	16.92	24.61	18.54	0.05	-11.53
PhysNet [20]	UBFC	10.24	16.54	12.46	0.29	-8.15
	PURE	13.22	19.61	14.73	0.23	-10.59
TS-CAN [21]	UBFC	14.01	21.04	15.48	0.20	-10.18
	PURE	13.94	21.61	15.14	0.20	-9.94
PhysFormer [28]	UBFC	12.10	17.79	15.41	0.17	-10.53
	PURE	14.57	20.71	16.73	0.15	-12.15
EfficientPhys [9]	UBFC	13.78	22.25	15.15	0.09	-9.13
	PURE	14.03	21.62	15.32	0.17	-9.95
Spiking-Phys [46]	UBFC	14.15	—	16.22	0.15	—
	PURE	14.57	—	16.55	0.14	—
RhythmMamba [13]	UBFC	10.63	17.14	12.14	0.34	-8.28
	PURE	10.44	16.70	12.25	0.36	**-8.18**
DualPhysMamba [12]	UBFC	—	—	—	—	—
	PURE	9.69	**15.47**	11.08	**0.44**	-8.71
Ours	UBFC	**9.69**	**16.31**	**11.76**	**0.41**	**-7.37**
	PURE	**9.56**	15.89	**11.06**	0.40	-8.20

short-term local variations, facilitating the learning of more robust and transferable rPPG representations.

5.3. Computational Cost

As shown in Table 7, we compare the number of parameters and multiply-accumulate operations (MACs) of end-to-end open-source methods, along with their performance on the PURE dataset. For fair comparison, all models were evaluated using the same toolkit with a unified input size of $160 \times 128 \times 128$ ($T \times H \times W$). Notably, our

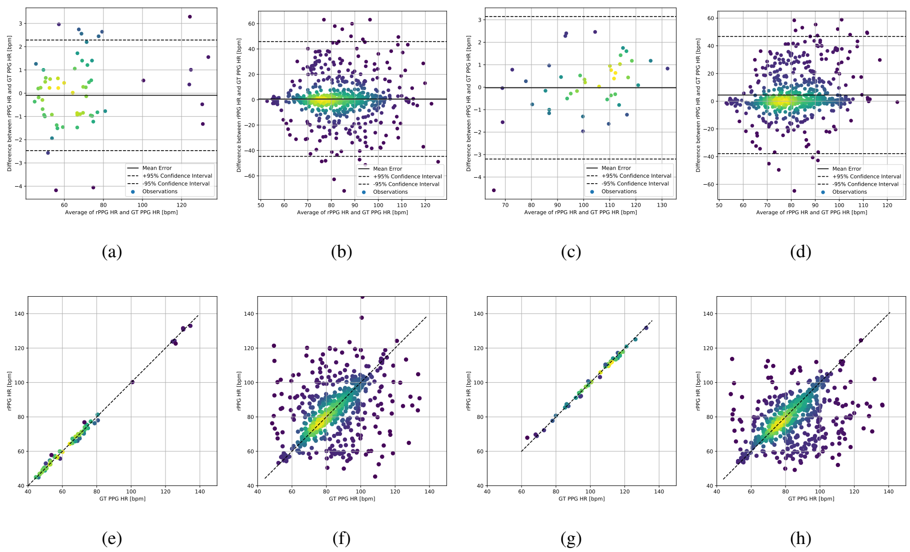


Figure 3: Cross-dataset evaluation results. Bland-Altman plots (a-d) and scatter plots (e-h) for models trained on UBFC and tested on PURE, trained on UBFC and tested on MMPD, trained on PURE and tested on UBFC, and trained on PURE and tested on MMPD.

Table 6: Dual-source cross-dataset evaluation by training on PURE and UBFC-rPPG and testing on MMPD.

Method	MAE↓	RMSE↓	ρ ↑
Green [43]	21.68	27.69	-0.01
PhysNet [20]	11.00	17.30	0.28
PhysFormer [28]	11.40	17.50	0.23
EfficientPhys [9]	11.80	18.90	0.22
RhythmFormer [14]	10.50	16.72	0.28
RhythmMamba [13]	12.03	18.65	0.31
PhysLLM [47]	9.95	**14.96**	–
PHASE-Net [48]	9.76	16.07	0.39
Ours	**9.62**	16.25	**0.42**

method achieves the lowest computational cost (7.37G MACs) among all compared deep learning approaches, representing a 43.0% reduction in MACs compared to the state-of-the-art DualPhysMamba [12], while simultaneously reducing the MAE from 0.18 to 0.15 on the PURE dataset. Although PhysMamba [7] has fewer parameters, its estimation error is substantially higher than that of our method. These results demonstrate that GLP-Mamba achieves a favorable trade-off between computational efficiency and physiological measurement accuracy, making it a promising solution for real-time health monitoring under resource-constrained conditions.

Table 7: Comparison of computational cost and performance.

Method	Param. (M)↓	MACs (G)↓	MAE↓	RMSE↓	ρ ↑
DeepPhys [19]	7.50	120.00	0.83	1.54	0.99
PhysNet [20]	0.70	70.12	2.10	2.60	0.99
TS-CAN [21]	7.50	120.00	2.48	9.01	0.92
PhysFormer [28]	7.38	50.60	1.10	1.75	0.99
EfficientPhys [9]	7.44	60.69	1.33	5.99	0.97
PhysMamba [7]	**0.56**	47.30	1.10	1.75	0.99
RhythmMamba [13]	2.00	13.24	0.23	0.34	0.99
MaKan-Mixer [11]	3.80	12.25	0.25	0.40	0.99
DualPhysMamba [12]	1.15	12.94	0.18	0.31	0.99
Ours	1.84	**7.37**	**0.15**	**0.29**	**0.99**

5.4. Ablation Study

We conduct detailed ablation studies on the MMPD dataset to investigate the contributions of each component in the DME and TPM and to analyze how they affect the overall performance.

Impact of Differential Motion Encoder. Table 8 presents the ablation results for DME components. Compared with the RGB-only baseline, using the short-difference branch reduces the MAE by 54.8% and the RMSE by 27.8%, while improving the correlation coefficient ρ by 52.3%. The long-difference branch achieves even larger gains, reducing the MAE by 59.1% and the RMSE by 36.4%, while improving ρ by 59.1%. These results indicate that temporal variation is much more informative than raw appearance information for pulse extraction on the challenging MMPD dataset. The stronger performance of the long-difference branch further suggests that a larger temporal interval is more effective for enhancing subtle physiological variations. Combining RGB with difference features leads to further improvements, showing that appearance information mainly serves as complementary spatial context. The best performance is achieved when all three branches are used together, demonstrating that spatial appearance, short-term temporal changes, and long-term temporal differences provide complementary information for robust rPPG estimation. In addition, adaptive fusion brings a further gain on top of this three-branch design. This finding is consistent with the analysis in Section 5.2, which shows that the DME improves cross-dataset performance by enhancing temporal pulse cues and reducing appearance interference.

Table 8: Ablation study of DME components on the MMPD dataset. Short denotes Short-Diff, Long denotes Long-Diff, and Fusion denotes Adaptive Fusion.

RGB	Short	Long	Fusion	MAE↓	RMSE↓	ρ ↑
✓	✗	✗	✗	11.70	15.48	0.44
✗	✓	✗	✗	5.29	11.17	0.67
✗	✗	✓	✗	4.79	9.84	0.70
✓	✓	✗	✓	3.68	8.02	0.80
✓	✗	✓	✓	3.49	7.51	0.83
✓	✓	✓	✗	3.32	7.08	0.85
✓	✓	✓	✓	**3.08**	**6.50**	**0.88**

Impact of Multi-temporal Scale. To further analyze the influence of temporal offsets in the DME, we evaluate different combinations of Short-Diff and Long-Diff branches on the MMPD dataset. As shown in Table 9, Short=1 and Long=3 achieves the best performance. When the Short offset is increased, the short-term branch deviates from instantaneous pulse fluctuations, reducing its sensitivity to fine-grained physiological cues. Using overly large offsets (e.g., Short=5 and Long=7) leads to notable performance degradation, as large temporal intervals introduce irregular motion variations that interfere with the adaptive fusion and weaken the spatiotemporal representations learned by DME. These results validate that Short=1 and Long=3 provides an optimal balance for capturing pulse-related temporal variations while suppressing motion artifacts.

Table 9: Ablation study of temporal offset combinations in the DME on the MMPD dataset. Short and Long denote the temporal offsets used in the Short-Diff and Long-Diff branches, respectively.

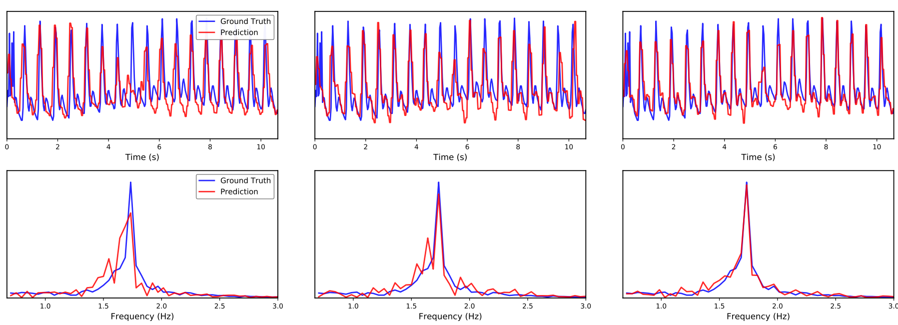
Short	Long	MAE↓	RMSE↓	$\rho \uparrow$
1	2	3.78	8.03	0.80
1	3(Ours)	**3.08**	**6.50**	**0.88**
1	5	3.61	7.81	0.81
1	7	3.67	7.65	0.82
2	3	3.55	7.88	0.82
2	5	3.40	7.45	0.83
2	7	3.49	7.54	0.82
3	5	3.68	7.81	0.82
3	7	3.53	7.49	0.82
5	7	4.24	8.92	0.75

Impact of Twin-Path Mamba. Table 10 shows the ablation results for TPM components. The global path significantly outperforms the local path, indicating that modeling long-range periodic dependencies is more critical than relying solely on short-term local dynamics on the challenging MMPD dataset. Combining both paths further improves performance, suggesting that local temporal variations provide complementary fine-grained cues to the global rhythmic representation. Notably, compared with the version without DAG, introducing DAG reduces the MAE by 14.4% and the RMSE by 16.8%, while also improving the correlation coefficient ρ to 0.88. This suggests that, under the large appearance and motion variations in MMPD, a fixed fusion strategy is insufficient, while DAG can dynamically adjust the contributions of global and local features to better accommodate distribution shifts.

To further investigate how different path configurations contribute to temporal modeling, we visualize their modeling results on the same sample in Fig. 4. Both single-path variants can capture the major signal trends, but each still shows slight deviations in either waveform detail or spectral response. In contrast, the full model produces predictions that better match the ground truth in both the time and frequency domains, suggesting that the interaction between the two pathways plays an important

Table 10: Ablation study of TPM components on the MMPD dataset. Global denotes the global Mamba branch, Local denotes the local convolutional branch.

Global	Local	DAG	MAE↓	RMSE↓	ρ ↑
✓	✗	✗	3.84	8.68	0.76
✗	✓	✗	5.06	10.98	0.64
✓	✓	✗	3.60	7.81	0.81
✓	✓	✓	**3.08**	**6.50**	**0.88**



(a) (b) (c)

Figure 4: Qualitative comparison of different path configurations on the same sample. (a) Local-only path. (b) Global-only path. (c) Full GLP-Mamba. Each subfigure shows time-domain signals (top) and frequency-domain spectra (bottom).

role.

Combining the quantitative metrics in Table 10 and the qualitative visualizations in Fig. 4, we further analyze the underlying principles of the TPM module, which reveals the distinct functional roles of the global and local pathways. The Local-only variant, while sensitive to transient dynamics, lacks long-range constraints on physiological rhythms, leading to periodic structure deviations manifested as waveform distortions and spectral peak shifts. Conversely, the Global-only variant effectively identifies the dominant heart rate frequency but struggles with fine-grained temporal alignment, often resulting in phase lags or subtle waveform inconsistencies. These observations indicate that single-path modeling alone is insufficient for capturing the dual-scale characteristics of rPPG signals, failing to balance rhythmic stability with local precision. These findings are consistent with our design motivation that effective rPPG estimation requires both long-range rhythm modeling and fine-grained local temporal perception.

Impact of Temporal Modeling Backbone. To verify the effectiveness of Mamba for temporal physiological modeling, we replace the temporal modeling backbone with TCN[49] and Transformer while keeping the other components unchanged. For a fair controlled comparison, the TCN and Transformer variants are designed with similar parameter scales to the original Mamba branch, avoiding large capacity gaps. As shown in Table 11, Mamba achieves the best performance in terms of MAE, RMSE, and ρ, indicating its stronger ability to capture pulse-related temporal dependencies. Although TCN obtains comparable results, it is still slightly inferior to Mamba across all three metrics. In contrast, Transformer shows a clear performance drop on the challenging MMPD dataset, suggesting that directly applying self-attention may be less robust under complex motion and illumination variations. These results demonstrate that Mamba is a more effective temporal modeling backbone for GLP-Mamba.

Table 11: Ablation study of different temporal modeling backbones on the MMPD dataset.

Backbone	MAE↓	RMSE↓	ρ ↑
TCN	3.15	6.54	0.86
Transformer	4.16	9.21	0.73
Mamba (Ours)	**3.08**	**6.50**	**0.88**

Impact of Network Depth. To examine the influence of network depth and justify the default setting of GLP-Mamba, we vary the number of GLP-Mamba blocks N while keeping other settings unchanged. In our main model, N is set to 24. As shown in Table 12, the performance improves as the network depth increases from $N = 6$ to $N = 24$, where the full GLP-Mamba achieves the best MAE, RMSE, and ρ values of 3.08, 6.50, and 0.88, respectively. This suggests that increasing the depth helps the model capture richer temporal physiological dependencies. However, when the depth is further increased to $N = 30$, the performance drops significantly, indicating that an overly deep architecture may introduce optimization difficulty or redundant temporal

Table 12: Ablation study on network depth (number of GLP-Mamba blocks N).

Depth	MAE↓	RMSE↓	ρ ↑
$N = 6$	3.40	7.35	0.83
$N = 12$	3.10	7.04	0.84
$N = 18$	3.10	6.87	0.85
$N = 24$(Ours)	**3.08**	**6.50**	**0.88**
$N = 30$	4.42	9.43	0.73

representations. Therefore, we adopt $N = 24$ as the default depth of GLP-Mamba in all main experiments.

Overall, these ablation studies validate the effectiveness of the main designs in GLP-Mamba. The DME-related results show that multi-scale temporal differences and adaptive fusion can better enhance pulse-related cues than RGB appearance alone. The TPM ablation further demonstrates that the global and local pathways are complementary, while DAG improves feature fusion under challenging motion and appearance variations. In addition, the backbone and depth analyses show that the Mamba-based temporal modeling branch and the default depth of $N = 24$ provide a better balance between representation capacity and robustness. These results together confirm the contribution of multi-scale motion encoding, global-local temporal modeling, and adaptive fusion to the final performance.

5.5. Comparison Under Challenging Scenarios

We further conduct a qualitative comparison on several challenging samples from the MMPD dataset to assess the robustness of the proposed method under complex interference. As shown in Fig. 5, we visualize the estimated rPPG signals of our method and RhythmMamba [13] together with the ground truth. In relatively difficult scenarios, especially those involving walking and head rotation, our method produces predictions that more closely follow the ground-truth waveform, with better temporal consistency and fewer visible distortions. Quantitatively, our method outperforms RhythmMamba across all three scenarios, achieving higher correlation and SNR in talking (ρ=0.86 vs. 0.85, SNR=2.26 vs. 1.11 dB), walking (ρ=0.77 vs. 0.24, SNR=−3.52 vs. −13.17 dB), and rotation (ρ=0.89 vs. 0.81, SNR=2.28 vs. 1.38 dB),

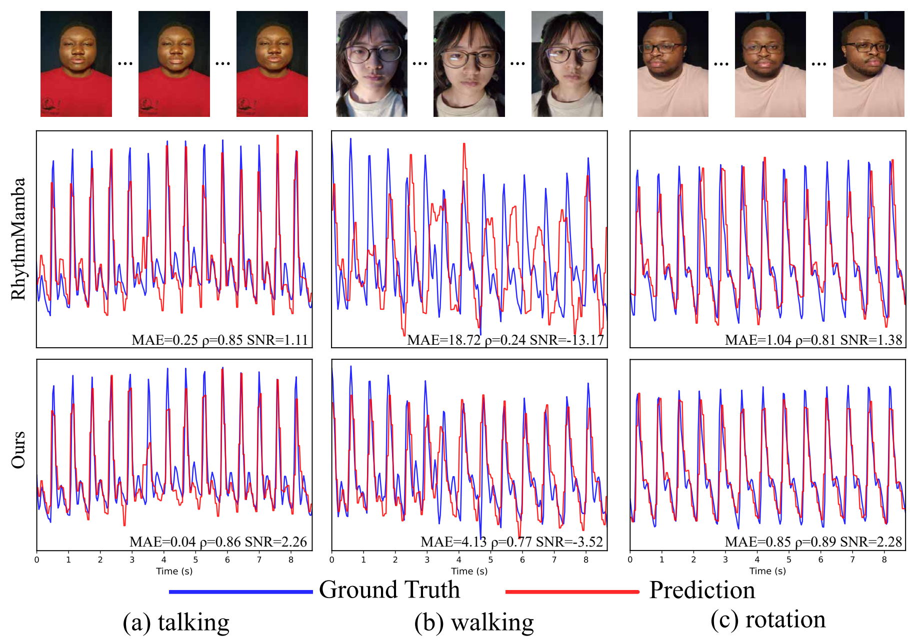


Figure 5: Comparison of prediction results for different challenging scenarios in the MMPD dataset. The blue and red curves denote the ground-truth and predicted rPPG signals, respectively. Quantitative metrics are reported for each displayed case to provide a more objective comparison.

demonstrating superior robustness under challenging motion conditions. These results provide qualitative evidence that the proposed method is more robust to motion interference and can maintain more stable rPPG estimation under challenging conditions. This observation is consistent with our design motivation.

6. Limitations

Although GLP-Mamba achieves promising performance on existing rPPG benchmarks, several aspects remain worthy of further investigation. First, although the current evaluation covers three widely used public rPPG datasets, the diversity of real-world acquisition conditions is still limited. Future work will further evaluate the model in more complex scenarios, such as low-light conditions, occlusions, and outdoor environments. Second, MMPD contains variations in motion, illumination, skin tone, and subject appearance, providing a useful benchmark for robustness evaluation.

However, more fine-grained subgroup analysis remains necessary to better understand the model behavior under different acquisition factors. Finally, DME adopts predefined temporal differences to enhance motion-related physiological cues. Although the temporal offset ablation supports the adopted setting, fixed temporal differences may limit adaptability when video conditions or physiological rhythms vary. Future work will explore more adaptive temporal-difference mechanisms and further compare them with existing rPPG motion representations.

7. Conclusion

In this paper, we propose GLP-Mamba to address the fundamental challenge of balancing long-term rhythm stability with short-term local sensitivity in rPPG estimation. The proposed architecture integrates a Differential Motion Encoder (DME) to enhance subtle pulse-related cues under complex spatiotemporal interference, and a Twin-Path Mamba (TPM) module that jointly models global physiological rhythms and local temporal variations. Extensive experiments on multiple public rPPG datasets demonstrate that GLP-Mamba achieves competitive performance, favorable cross-dataset generalization, and a clear balance between estimation accuracy and computational efficiency. Future work will focus on improving its resistance to severe motion artifacts and exploring parameter compression strategies for more efficient deployment without compromising estimation accuracy.

Data availability

The data supporting the findings of this study are available from the authors upon reasonable request.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

We used Generative AI tools, such as ChatGPT, only to improve the language clarity and readability of this manuscript. We carefully reviewed and revised the generated text where necessary and take full responsibility for the final content of the manuscript.

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References

[1] J. Shi, I. Alikhani, X. Li, Z. Yu, T. Seppänen, G. Zhao, Atrial fibrillation detection from face videos by fusing subtle variations, IEEE Transactions on Circuits and Systems for Video Technology 30 (8) (2019) 2781–2795.

[2] B. Huang, S. Hu, Z. Liu, C.-L. Lin, J. Su, C. Zhao, L. Wang, W. Wang, Challenges and prospects of visual contactless physiological monitoring in clinical study, NPJ Digital Medicine 6 (1) (2023) 231.

[3] M.-Z. Poh, D. J. McDuff, R. W. Picard, Non-contact, automated cardiac pulse measurements using video imaging and blind source separation., Optics express 18 (10) (2010) 10762–10774.

[4] G. De Haan, V. Jeanne, Robust pulse rate from chrominance-based rppg, IEEE transactions on biomedical engineering 60 (10) (2013) 2878–2886.

[5] W. Wang, A. C. Den Brinker, S. Stuijk, G. De Haan, Algorithmic principles of remote ppg, IEEE Transactions on Biomedical Engineering 64 (7) (2016) 1479–1491.

[6] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, in: First conference on language modeling, 2024.

[7] C. Luo, Y. Xie, Z. Yu, Physmamba: Efficient remote physiological measurement with slowfast temporal difference mamba, in: Chinese Conference on Biometric Recognition, Springer, 2024, pp. 248–259.

[8] J. S. Lee, G. Hwang, M. Ryu, S. J. Lee, Lstc-rppg: Long short-term convolutional network for remote photoplethysmography, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 6015–6023.

[9] X. Liu, B. Hill, Z. Jiang, S. Patel, D. McDuff, Efficientphys: Enabling simple, fast and accurate camera-based cardiac measurement, in: Proceedings of the IEEE/CVF winter conference on applications of computer vision, 2023, pp. 5008–5017.

[10] M. Hu, F. Qian, D. Guo, X. Wang, L. He, F. Ren, Eta-rppgnet: Effective time-domain attention network for remote heart rate measurement, IEEE Transactions on Instrumentation and Measurement 70 (2021) 1–12.

[11] H. Zhang, F. Liao, G. Yuan, H. Jin, B. Xie, X. Cao, M. Fu, J. Zheng, Makan-mixer: Channel interaction-based mamba method for rppg extraction, IEEE Journal of Biomedical and Health Informatics (2025).

[12] J. Li, J. Peng, Dualphysmamba: Efficient end-to-end network based on noise modeling for remote photoplethysmography measurement, Applied Soft Computing (2025) 114328.

[13] B. Zou, Z. Guo, X. Hu, H. Ma, Rhythmmamba: Fast, lightweight, and accurate remote physiological measurement, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 39, 2025, pp. 11077–11085.

[14] B. Zou, Z. Guo, J. Chen, J. Zhuo, W. Huang, H. Ma, Rhythmformer: Extracting patterned rppg signals based on periodic sparse attention, Pattern Recognition 164 (2025) 111511.

[15] Z. Ji, B. Zou, X. Kui, S. Thureau, S. Ruan, Global and local mamba network for multi-modality medical image super-resolution, Pattern Recognition (2025) 112888.

[16] G.-B. Truong, T.-T. Tran, N.-L. Than, V. Q. Nguyen, T. H. Nguyen, V.-T. Pham, Sc-mambafew: Few-shot learning based on mamba and selective spatial-channel attention for bearing fault diagnosis, Computers and Electrical Engineering 123 (2025) 110004.

[17] K. Liang, Z. Lin, J. Yang, T. Yu, J. Chi, F. Liu, Gl-mamba-net: A magnetic resonance imaging restoration network with global-local mamba, Magnetic Resonance Imaging (2025) 110574.

[18] M. Lewandowska, J. Rumiński, T. Kocejko, J. Nowak, Measuring pulse rate with a webcam—a non-contact method for evaluating cardiac activity, in: 2011 federated conference on computer science and information systems (FedCSIS), IEEE, 2011, pp. 405–410.

[19] W. Chen, D. McDuff, Deepphys: Video-based physiological measurement using convolutional attention networks, in: Proceedings of the european conference on computer vision (ECCV), 2018, pp. 349–365.

[20] Z. Yu, X. Li, G. Zhao, Remote photoplethysmograph signal measurement from facial videos using spatio-temporal networks, in: Proceedings of the British Machine Vision Conference (BMVC), 2019, bMVC 2019, Cardiff, UK, Sep 9–12, 2019. Paper 277.

[21] X. Liu, J. Fromm, S. Patel, D. McDuff, Multi-task temporal shift attention networks for on-device contactless vitals measurement, Advances in Neural Information Processing Systems 33 (2020) 19400–19411.

[22] J. Li, J. Cheng, R. Song, Y. Liu, Lst-rppg: A long-range spatio-temporal model for high-accuracy heart rate variability measurement, Expert Systems with Applications (2025) 129526.

[23] X. Zhang, F. Qiao, Q. Zang, Z. Chen, X. Xie, S. Zhao, S. Lin, Y. Li, Noise-resilient spatial–channel refinement for robust remote photoplethysmography, Frontiers in Bioengineering and Biotechnology 14 (2026) 1819324.

[24] C. Zhiyuan, H. Wang, Y. Han, J. Lv, B. Li, Phys-vim: State space model for remote physiological measurement, Neurocomputing (2025) 131405.

[25] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, M. Tegmark, Kan: Kolmogorov-arnold networks, arXiv preprint arXiv:2404.19756 (2024).

[26] S. Fan, H. Wang, F. Zhang, Fsmamba: A dual-expert architecture with fast global attention and local-enhanced state-space mamba for time series forecasting, Knowledge-Based Systems (2025) 115233.

[27] S. Fu, P. Liu, T. Zhang, H. Zhao, Q. Zhou, Synergistic mamba: Mastering global frequency and local spatial contexts for low-light image enhancement, Expert Systems with Applications (2025) 130946.

[28] Z. Yu, Y. Shen, J. Shi, H. Zhao, P. H. Torr, G. Zhao, Physformer: Facial video-based physiological measurement with temporal difference transformer, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 4186–4196.

[29] Z. Yu, Y. Shen, J. Shi, H. Zhao, Y. Cui, J. Zhang, P. Torr, G. Zhao, Physformer++: Facial video-based physiological measurement with slowfast temporal difference transformer, International Journal of Computer Vision 131 (6) (2023) 1307–1330.

[30] Z. Yu, X. Li, X. Niu, J. Shi, G. Zhao, Autohr: A strong end-to-end baseline for remote heart rate measurement with neural searching, IEEE Signal Processing Letters 27 (2020) 1245–1249.

[31] R. Stricker, S. Müller, H.-M. Gross, Non-contact video-based pulse rate measurement on a mobile service robot, in: The 23rd IEEE International Symposium on Robot and Human Interactive Communication, IEEE, 2014, pp. 1056–1062.

[32] S. Bobbia, R. Macwan, Y. Benezeth, A. Mansouri, J. Dubois, Unsupervised skin tissue segmentation for remote photoplethysmography, Pattern recognition letters 124 (2019) 82–90.

[33] J. Tang, K. Chen, Y. Wang, Y. Shi, S. Patel, D. McDuff, X. Liu, Mmpd: Multi-domain mobile video physiology dataset, in: 2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, 2023, pp. 1–5.

[34] X. Liu, G. Narayanswamy, A. Paruchuri, X. Zhang, J. Tang, Y. Zhang, R. Sengupta, S. Patel, Y. Wang, D. McDuff, rppg-toolbox: Deep remote ppg toolbox, Advances in Neural Information Processing Systems 36 (2023) 68485–68510.

[35] R. Špetlík, V. Franc, J. Matas, Visual heart rate estimation with convolutional neural network, in: Proceedings of the british machine vision conference, Newcastle, UK, 2018, pp. 3–6.

[36] H. Lu, H. Han, S. K. Zhou, Dual-gan: Joint bvp and noise modeling for remote physiological measurement, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2021, pp. 12404–12413.

[37] R. Song, H. Chen, J. Cheng, C. Li, Y. Liu, X. Chen, Pulsegan: Learning to generate realistic pulse waveforms in remote photoplethysmography, IEEE Journal of Biomedical and Health Informatics 25 (5) (2021) 1373–1384.

[38] X. Niu, H. Han, S. Shan, X. Chen, Synrhythm: Learning a deep heart rate estimator from general to specific, in: 2018 24th international conference on pattern recognition (ICPR), IEEE, 2018, pp. 3580–3585.

[39] E. Lee, E. Chen, C.-Y. Lee, Meta-rppg: Remote heart rate estimation using a transductive meta-learner, in: European conference on computer vision, Springer, 2020, pp. 392–409.

[40] S.-Q. Liu, P. C. Yuen, A general remote photoplethysmography estimator with spatiotemporal convolutional network, in: 2020 15th IEEE international con-

ference on automatic face and gesture recognition (FG 2020), IEEE, 2020, pp. 481–488.

[41] J. Li, Z. Yu, J. Shi, Learning motion-robust remote photoplethysmography through arbitrary resolution videos, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 37, 2023, pp. 1334–1342.

[42] J. Speth, N. Vance, P. Flynn, A. Czajka, Non-contrastive unsupervised learning of physiological signals from video, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 14464–14474.

[43] W. Verkruysse, L. O. Svaasand, J. S. Nelson, Remote plethysmographic imaging using ambient light., Optics express 16 (26) (2008) 21434–21445.

[44] C. S. Pilz, S. Zaunseder, J. Krajewski, V. Blazek, Local group invariance for heart rate estimation from face videos in the wild, in: Proceedings of the IEEE conference on computer vision and pattern recognition workshops, 2018, pp. 1254–1262.

[45] G. De Haan, A. Van Leest, Improved motion robustness of remote-ppg by using the blood volume pulse signature, Physiological measurement 35 (9) (2014) 1913.

[46] M. Liu, J. Tang, Y. Chen, H. Li, J. Qi, S. Li, K. Wang, J. Gan, Y. Wang, H. Chen, Spiking-physformer: Camera-based remote photoplethysmography with parallel spike-driven transformer, Neural Networks 185 (2025) 107128.

[47] Y. Xie, B. Zhao, M. Dai, J.-P. Zhou, Y. Sun, T. Tan, W. Xie, L. Shen, Z. Yu, Physllm: Harnessing large language models for cross-modal remote physiological sensing, arXiv preprint arXiv:2505.03621 (2025).

[48] B. Zhao, D. Guo, J. Cao, Y. Xu, B. Zou, T. Tan, Y. Sun, Z. Yu, Phase-net: Physics-grounded harmonic attention system for efficient remote photoplethysmography measurement, arXiv preprint arXiv:2509.24850 (2025).

[49] S. Bai, J. Z. Kolter, V. Koltun, An empirical evaluation of generic convolutional and recurrent networks for sequence modeling, arXiv preprint arXiv:1803.01271 (2018).